Research statement

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My research field is mathematical physics. I completed a PhD entitled "Semi-classical limits of 2D fermions under high magnetic fields" supervised by Nicolas Rougerie at the ENS Lyon. Passionate about teaching mathematics, I am interested in the analysis of many-body systems of interacting quantum particles, with the aim of deriving effective theories, in particular in mean-field and semi-classical limits. These results provide approximate descriptions of quantum systems using theories that are relevant in theoretical and numerical physics.

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I - General Introduction

I.1 - N-body Quantum Physics

The quantum N-body problem corresponds is the study of systems of N interacting particles governed by the laws of quantum mechanics. Typically, to goal is to compute the energy of the system, its particle density, or its dynamics. In quantum mechanics, a particle placed in a domain $\Omega \subseteq \mathbb{R}^d$ is described by a vector in the Hilbert space $L^2(\Omega, \mathbb{C})$. A system of N particles is described by an element of the tensor product space

$$L^2(\Omega, \mathbb{C})^{\otimes N} \cong L^2(\Omega^N, \mathbb{C}).$$

The energy of a system is described by a self-adjoint operator acting on the Hilbert space of states, called the Hamiltonian, here denoted by $\mathcal{H}$. When measuring the energy of a quantum system, the result is an eigenvalue of the Hamiltonian, and the state is projected onto the associated eigenspace. Given a physical state ψ, the particle density corresponds to the squared modulus of this state, and is therefore integrable. The dynamics are governed by the Schrödinger equation:

$$i\hbar\partial_t\psi = \mathcal{H}\psi.$$

$\hbar$ is the reduced Planck constant, which sets the scale of quantum phenomena. The Hamiltonian of a system of N spinless particles takes the form:

$$\mathcal{H}_N := \sum_{i=1}^{N} \left(-\hbar^2\Delta_i + V(x_i)\right) + \sum_{i<j} w(x_i - x_j). \tag{I.1}$$

The operator $-\Delta$ corresponds to the kinetic energy of the system. $V : \Omega \to \mathbb{R}$ is the external potential acting on the particles. For instance, V may model an electrostatic potential. w represents the interaction potential between a pair of particles. The potentials act multiplicatively on the physical states. The notations $\Delta_i, V(x_i)$ indicate that these operators act only on the ith particle, i.e. on the ith coordinate of $L^2(\Omega^N, \mathbb{C})$. Similarly, $w(x_i - x_j)$ acts on the pair of particles (i, j), that is, on the ith and j^{th} coordinates of $L^2(\Omega^N, \mathbb{C})$. The N-body Hamiltonian (I.1) thus consists of the sum of the kinetic and potential energies of each particle, together with the sum of the interactions between each pair of particles.

In a N-body state $\psi_N \in L^2(\Omega^N, \mathbb{C})$, the particle densities are correlated if ψ_N is not factorized in the form

$$\psi_N = u^{\otimes N}, \text{ with } u \in L^2(\Omega, \mathbb{C}).$$

Solving the quantum N-body problem amounts to diagonalizing the Hamiltonian (I.1). This is in general extremely difficult due to interactions and correlations between particles. Even numerically, the study of this problem quickly becomes infeasible for relevant values of N. Indeed, a numerical method must begin by using a finite-dimensional approximation of the Hilbert space. $L^2(\Omega, \mathbb{C})$ is then replaced by $\mathbb{C}^m$, but the N-particle space becomes $\mathbb{C}^{m^N}$. Since the dimension depends exponentially on the number of particles, the direct numerical treatment of quantum N-body models is currently impossible. It is therefore necessary to resort to approximate models that still capture the essential quantum phenomena.

I.2 - Mean-field regime and semi-classical limit

Our goal is to obtain effective theories, simpler and more amenable to numerical analysis. For example, density functional theory is a class of effective models that is widely used. Instead of minimizing over the space $L^2(\Omega, \mathbb{C})$, one can then minimize a functional that depends only on the density. Mathematical results

on density functional theories are known to be useful in numerical methods for calibrating free parameters. Density functional theories are the main tools for quantum numerical computations, for instance in quantum chemistry or condensed matter physics, where one aims to compute the electronic density.

To obtain effective theories, one of the main tools is the mean-field approximation. This amounts to assuming that particles are not correlated, that is, each particle feels the same mean interaction field generated by the gas of particles. One can expect this approximation to be accurate for large systems, since the interaction between a pair of particles becomes negligible compared to the mean interaction field.

Since physical particles are indistinguishable, the particle density must be invariant under coordinates permutation. There are two canonical ways to impose this symmetry on $L^2(\Omega^N, \mathbb{C})$, allowing one to model bosons or fermions. Indeed, a function that is symmetric or antisymmetric under the exchange of two coordinates is respectively called bosonic or fermionic.

When taking the limit $N \to \infty$, in order to obtain a meaningful limit, one must ensure that all terms in the Hamiltonian (I.1) are of the same order of magnitude. To this end, one divides the interaction term by N:

$$\mathcal{H}_N := \sum_{i=1}^{N} \left(-\hbar^2 \Delta_i + V(x_i)\right) + \frac{1}{N} \sum_{i<j} w(x_i - x_j) \tag{I.2}$$

and one sets $\hbar = 1$ for bosons or $\hbar = N^{-\frac{1}{d}}$ for fermions [28]. This fermionic peculiarity comes from the Pauli principle, which makes the kinetic energy of fermionic systems larger than that of a bosonic systems. For fermions, the mean-field regime is therefore coupled with a semi-classical regime $\hbar \to 0$. Indeed, by considering a limit where the scale of quantum phenomena goes to 0, one hopes to obtain the corresponding classical theory. This is called the correspondence principle and was first introduced by Niels Bohr [3].

II - Results and perspectives

II.1 - Mean-field under strong magnetic field: static

During my PhD [25], I studied two dimensional systems of fermions in the presence of a magnetic field. The N-body Hamiltonian is then

$$\mathcal{H}_N := \sum_{i=1}^{N} \left((-i\hbar\nabla_i + bA(x_i))^2 + V(x_i)\right) + \frac{1}{N} \sum_{i<j} w(x_i - x_j).$$

The magnetic Laplacian includes a vector potential A generating a constant, homogeneous magnetic field transverse to the plane where the particles are confined:

$$\nabla \wedge A = (0, 0, 1).$$

In a first result [7], we study the mean-field limit, coupled with a semi-classical and strong magnetic field limit, for the ground state of the fermionic system. In classical mechanics, charged particles subject to a homogeneous transverse magnetic field follow orbital trajectories. In quantum mechanics, the kinetic energy is quantized into discrete energy levels called Landau levels, separated by a constant gap. The main results of this article extend works by Lieb-Solovej-Yngvason [14] and Fournais-Madsen [24] by considering a regime where the gap between Landau levels is the dominant energy scale compared to interactions. Moreover, the particles are placed in a bounded domain, allowing the degeneracy of Landau levels to be finite. We thus obtain a limiting model where an arbitrary number of Landau levels are filled. Existing works, using confinement by a trapping potential, cannot describe this situation. We also discuss the physics of the limiting model in the last partially filled Landau level.

In this result, the regularity assumptions on the potentials are not optimal. We expect that this result remains valid if the potentials have a positive part in L^1 and a negative part in L^2. Under these assumptions, one must prove that the particles do not concentrate in the positive L^1 singularities of the potentials. This was achieved in [15] for the Coulomb potential.

Replacing the magnetic Laplacian by the Dirac operator would allow for a description of graphene. Indeed, in graphene, the dispersion relation at the Dirac cones in the band structure makes electron transport relativistic. Graphene is known to be a good material for the experimental observation of the quantum Hall effect [23]. It is therefore interesting to consider a model for $2d$ fermions within a framework of special relativity. Since the Dirac operator is not positive, one would assume the Dirac sea hypothesis, stating that in the vacuum all negative energy states are occupied. A difficulty for this result would be to handle general potentials: indeed, this would require adapting a Lieb-Thirring inequality for the Dirac operator with periodic magnetic boundary conditions.

Another interesting direction to explore would be to consider the case of a non-homogeneous (still transverse) magnetic field. By choosing a sufficiently regular magnetic field, at the scale l_b, the magnetic field behaves locally like a constant. Thus, the kinetic energy remains locally quantized into Landau levels. In the longer term, this research direction could open the way to the study of mean-field limits for systems of anyons, particles with intermediate statistics between bosons and fermions. Indeed, the Hamiltonian describing a system of anyons can be rewritten as a bosonic or fermionic Hamiltonian coupled to a highly singular and non-homogeneous magnetic field, in which each particle carries a magnetic flux. In the limit $N \to \infty$, this model naturally leads to a strong magnetic field regime, since the number of fluxes is proportional to the number of particles. This therefore offers the perspective of analysing anyonic systems beyond the quasi-bosonic or quasi-fermionic regimes usually considered [11]. As a first step, we would consider generalized anyons by regularizing the singularities of the magnetic field.

II.2 - Mean-field under strong magnetic field: dynamics

In the same context as the previous section, we also studied in [22] the dynamics of the system, starting from the Hartree equation for the one-particle density matrix:

$$i\hbar\partial_t\gamma = \left[(i\hbar\nabla + bA)^2 + V + w \star \rho_\gamma, \gamma\right],$$

where ρ_γ is the physical density. It is well known [2] [20] that this equation can be derived as a mean-field approximation of the N-body Schrödinger dynamics for interacting fermions. Here, we study the strong magnetic field limit, coupled with a semi-classical limit, and prove that the density converges to a solution of a gyrokinetic transport equation:

$$\partial_t\rho(t,x) + \nabla_x^\perp \left(V + w \star \rho\right)(t,x) \cdot \nabla_x\rho(t,x) = 0. \tag{II.1}$$

In a classical mechanics framework, works by Golse-St-Raymond [19] obtain similar results starting from the Vlasov equation. More recently, a work in the semi-classical limit by Ben Porat [12] treats the case where the gap between Landau levels is much smaller than the interactions. The main novelty of our result is to consider a quantum regime where the gap between Landau levels is of the same order of magnitude as the interaction energy. The proof consists in constructing a semi-classical density whose time evolution is obtained and then compared to the target equation.

A significant improvement of this result would be to reduce the regularity assumptions on the potentials. To this end, one could use methods similar to [5]. This would require establishing the equation for the dynamics within a Landau level, in order to directly control the equation for the Husimi function rather than the physical density. For the dynamics of the Husimi function, one expects to obtain, in addition to the gyrokinetic equation, a jump term between Landau levels. One would thus obtain a system of coupled equations with one equation for each Landau level. This would make it possible to construct a quantum

object, analogous to the Weyl quantization of the density on phase space $\mathbb{R}^d \times \mathbb{R}^d$, to be compared with the density matrix solution of the Hartree equation. Moreover, obtaining propagation of higher-order moments of the kinetic energy would facilitate the control of errors in the dynamics of the first reduced density. The propagation of certain commutators, between γ_b and the kinetic energy or moments of the position operator, could also be useful. These methods should also allow for improving the convergence rate in our theorem.

An ongoing related project, which I am conducting independently, is to take the N-body Schrödinger equation as a starting point instead of the mean-field dynamics to study the joint triple limit: mean-field, semi-classical, and strong magnetic field. The derivation of the Vlasov equation from a quantum N-body system is well known in a coupled mean-field and semi-classical regime [17, 18, 16]. However, deriving (II.1) from the N-body Schrödinger equation would complement the existing literature on this topic.

II.3 - Mean-field in large dimensions: dynamics

This section is devoted to the research direction corresponding to the works carried out during my postdoc [27] [26]. The first result, concerning the dynamics, is relatively easy to state, so we provide below the definitions needed to give a simplified version of the theorem.

Our goal is to justify the mean-field approximation in the limit where the dimension of space goes to infinity, rather than in the usual limit of a large number of particles in the mean-field regime described in (I.2). Indeed, in the physics literature, it is often stated that mean-field theories become exact when $d = +\infty$.

To this end, we consider the Bose-Hubbard model, both for its mathematical simplicity as a model on a finite graph, and for its well-known relevance in physics to describe a gas of strongly and locally interacting bosons [10]. One of the greatest successes of the Bose-Hubbard model is the description of a phase transition between a Mott insulator and a superfluid [13]. The usefulness of the limit $d \to \infty$ is motivated by the fact that the resulting mean-field theory already provides quantitatively accurate results in dimension $d = 3$ [8].

The mean-field limit obtained in [27] allows us to describe a phase transition starting from a quantum N-body model, which is generally not the case in the usual $N \to \infty$ limit. Our result concerns the dynamics of the system, which is thus approximated by that of the mean-field model. Another advantage of the regime $d \to \infty$ is that it allows one to treat strongly interacting particles. Indeed, it is no longer necessary to multiply the interaction term by the mean-field factor N^{-1}, which amounts to considering weak coupling between particles.

Let $\Lambda := (\mathbb{Z}/L\mathbb{Z})^d$ be the d-dimensional cubic lattice with periodic boundary conditions. To each site is associated a Hilbert space $\ell^2(\mathbb{C})$ with canonical basis $|n\rangle := (0, \dots, 0, \underbrace{1}_{n^{\text{th}} \text{ index}}, 0, \dots), n \in \mathbb{N}$.

We define the creation operator $a^\dagger$ and annihilation operator a by

$$\begin{aligned} &a\,|0\rangle := 0 \quad \forall n \in \mathbb{N}^*,\ a\,|n\rangle := \sqrt{n}\,|n-1\rangle\,, \\ &\forall n \in \mathbb{N},\ a^\dagger\,|n\rangle := \sqrt{n+1}\,|n+1\rangle\,, \end{aligned}$$

as well as the particle number operator $\mathcal{N} := a^\dagger a$. The Fock space is then

$$\mathcal{F} := \ell^2(\mathbb{C})^{\otimes|\Lambda|} \cong \mathcal{F}_+\left(L^2(\Lambda, \mathbb{C})\right) := \bigoplus_{n\in\mathbb{N}} L^2(\Lambda, \mathbb{C})^{\otimes_+ n}.$$

If A is an operator on $\ell^2(\mathbb{C})$ and $x \in \Lambda$, we denote by A_x the operator on $\mathcal{F}$ acting on site x as A and as the identity on the other sites.

The Bose-Hubbard Hamiltonian is

$$H_d := -\frac{J}{2d} \overbrace{\sum_{\substack{x,y\in\lambda \\ x\sim y}} a_x^\dagger a_y}^{\mathcal{O}(2d|\Lambda|)} + (J-\mu)\sum_{x\in\Lambda}\mathcal{N}_x + \frac{U}{2}\sum_{x\in\Lambda}\mathcal{N}_x(\mathcal{N}_x - 1).$$

where $J \in \mathbb{R}$ is the hopping amplitude (between nearest neighbours), $\mu \in \mathbb{R}$ the chemical potential, and U the on-site pair-interaction strength. It is important to note that the roles of the interaction term and the kinetic energy are reversed compared to the usual $N \to \infty$ limit. Indeed, for our scaling $d \to \infty$, the kinetic term acts on two sites, whereas interactions between particles occur only on the same site. It is therefore the hopping term that must be approximated by a mean-field method, not the interaction. The hopping amplitude must thus be divided by the coordination number of the graph, $2d$, to obtain terms of the same order of magnitude.

The associated dynamics can be written for a density matrix $\gamma_d \in L^\infty(\mathbb{R}_+, \mathcal{L}^1(\mathcal{F}))$:as

$$i\partial_t \gamma_d(t) = [H_d, \gamma_d(t)]\,. \tag{B-H}$$

where $\mathcal{L}^1(\mathcal{F})$ denotes the trace class on $\mathcal{F}$.

Finally, we define the reduced one-site density:

$$\gamma_d^{(1)} := \frac{1}{|\Lambda|}\sum_{x\in\Lambda} \mathrm{Tr}_{\Lambda\setminus\{x\}}(\gamma_d)\,.$$

The mean-field theory obtained in the limit $d \to \infty$ is thus a theory on a single lattice site. The Hamiltonian is given, for a state $\varphi \in \ell^2(\mathbb{C})$, by

$$h^\varphi := -J\big(\overline{\alpha_\varphi} a + \alpha_\varphi a^\dagger - |\alpha_\varphi|^2\big) + (J-\mu)\mathcal{N} + \frac{U}{2}\mathcal{N}(\mathcal{N}-1).$$

with the order parameter

$$\alpha_\varphi := \langle \varphi, a\varphi\rangle\,.$$

This theory exhibits a phase transition between a Mott insulator (MI) when $\alpha_\varphi = 0$ and a superfluid (SF) when $\alpha_\varphi > 0$.

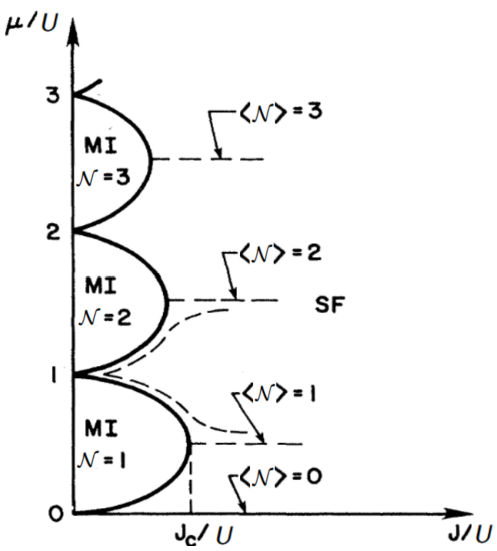

Figure 1 – Phase diagram of the Mott insulator (MI) / superfluid (SF) [8], obtained by minimizing the mean-field energy functional:

$$\langle \varphi, h_\varphi \varphi\rangle = -J|\alpha_\varphi|^2 + (J-\mu)\langle \varphi, \mathcal{N}\varphi\rangle + \tfrac{U}{2}\langle \varphi, \mathcal{N}(\mathcal{N}-1)\varphi\rangle\,.$$

Given a ground state $\varphi(J,\mu,U)$, the Mott insulator phase is defined as the region where $\alpha_\varphi(J,\mu,U) = 0$.

The associated dynamics is

$$i\partial_t \varphi(t) = h^{\varphi(t)}\varphi(t). \tag{mf}$$

With the additional notation

$$p_\varphi \coloneqq |\varphi\rangle\langle\varphi| \quad q_\varphi \coloneqq 1 - p_\varphi,$$

we can state the main result of [27]:

Theorem II.1: *S.Farhat D.P S.Petrat 2025*

If

— γ_d is a solution of (B-H) with $\gamma_d(0) \in \mathcal{L}^1(\mathcal{F})$ such that $\mathrm{Tr}(\gamma_d(0)) = 1$

— φ is a solution of (mf) with $\varphi(0) \in \ell^2(\mathbb{C})$ such that $\|\varphi\|_{\ell^2} = 1$

— $\exists c_1, c_2 > 0$ such that $\forall n \in \mathbb{N}$,

$$\mathrm{Tr}\left(p_\varphi(0)\mathbb{1}_{\mathcal{N}=n}\right) \leqslant c_1 e^{-\frac{n}{c_2}}, \quad \mathrm{Tr}\left(\gamma_d^{(1)}(0)\mathbb{1}_{\mathcal{N}=n}\right) \leqslant c_1 e^{-\frac{n}{c_2}}.$$

Then $\exists\ C \coloneqq C\left(J, c_1, c_2, \mathrm{Tr}\left(p_\varphi(0)\mathcal{N}\right)\right) > 0$ such that $\forall t \in \mathbb{R}_+$,

$$\left\|\gamma_d^{(1)}(t) - p_\varphi(t)\right\|_{\mathcal{L}^1} \leqslant e^{te^{C(t+1)}\sqrt{\ln(d)}}\left(\left\|\gamma_d^{(1)}(0) - p_\varphi(0)\right\|_{\mathcal{L}^1} + \frac{1}{d\sqrt{\ln(d)}}\right)$$

This theorem allows us to propagate the difference between the dynamics given by the Bose-Hubbard model and that of the mean-field model: if $\left\|\gamma_d^{(1)}(0) - p_\varphi(0)\right\|_{\mathcal{L}^1} = \mathcal{O}\left(\frac{1}{d}\right)$, then $\forall t \in \mathbb{R}_+$,

$$\left\|\gamma_d^{(1)}(t) - p_\varphi(t)\right\|_{\mathcal{L}^1} \leqslant 2e^{te^{C(t+1)}\sqrt{\ln(d)} - \ln(d)} \underset{d\to\infty}{\to} 0.$$

Another consequence is the convergence of the order parameter:

$$\left\|\left(\gamma_d^{(1)} - p_\varphi\right)a\right\|_{\mathcal{L}^1} \underset{d\to\infty}{\to} 0.$$

In the article, we also prove the existence and uniqueness of the solution to the mean-field dynamics.

In the longer term, we would like to refine the dependence on d in our estimate, ideally obtaining results that are already useful in dimension $d = 3$. To move in this direction, we plan to compute and incorporate corrections to the mean-field dynamics by taking quantum fluctuations into account, with the aim of proving a bound involving higher powers of d^{-1}. This project is based on quantum perturbation theory and tools related to the Bogolyubov approximation, which are developed in particular in [4].

Another natural objective would be to extend this approach to the Fermi-Hubbard model, or to treat the case of an infinite graph based on [9] in order to define the dynamics on the infinite lattice.

II.4 - Mean-field in the large coordination limit: static

In a more recent result [26], we study the analogue of Theorem II.1 for the ground state energy. The mathematical setting is here generalized to an arbitrary graph with homogeneous coordination (number of nearest neighbours) z. We investigate the ground state energy of the Bose–Hubbard Hamiltonian in the limit where $z \to \infty$. Note that for Theorem II.1, $z = 2d$. The main tool is a quantum de Finetti theorem (see [21] for a review on this topic), used to obtain the lower bound on the energy. After reducing the model to a local model consisting of a site of the graph and its shell of nearest neighbours, the problem turns out to be mathematically identical to that of an impurity evolving in a bosonic field. Depending on the models considered, this particle is commonly called a *tracer particle* or a *polaron*. We thus develop a new polaron-type version of a quantum de Finetti theorem, which constitutes an extension of existing

results in quantum de Finetti theory. Our theorem covers the case where the Hilbert space is a tensor product of a Hilbert space with a bosonic Fock space. This theorem is then applied to the study of the convergence of the ground state energy of the Bose–Hubbard model after reducing it to a polaron-type model. We are currently working on extending the main result of [26] in order to obtain the convergence of the ground state, as well as an estimate of the rate of convergence.

With the de Finetti theorem proved in [26], we hope to complement existing works on polaron models, in particular [6], by treating a model including an external potential acting both on the impurity and on the bosonic field, which, to my knowledge, is new in the field.

Another line of research consists in considering variations of the Bose–Hubbard model, for instance by considering a hopping amplitude that depends on the distance in the graph for greater physical realism, or by including an inhomogeneous external potential. In the latter case, we expect that factorization remains valid in the limit of large coordination, but the resulting mean-field theory would then depend on the position in the graph. In the presence of disorder, the physics is very rich as the Bose–Hubbard is known to exhibit a third phase called Bose Glass.

Finally, after having treated the dynamics and the ground state energy, we naturally plan to study the excited states, as well as the convergence of Gibbs measures by adapting [1] to the regime $z \to \infty$ in order to extend the description to non-zero temperatures.

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